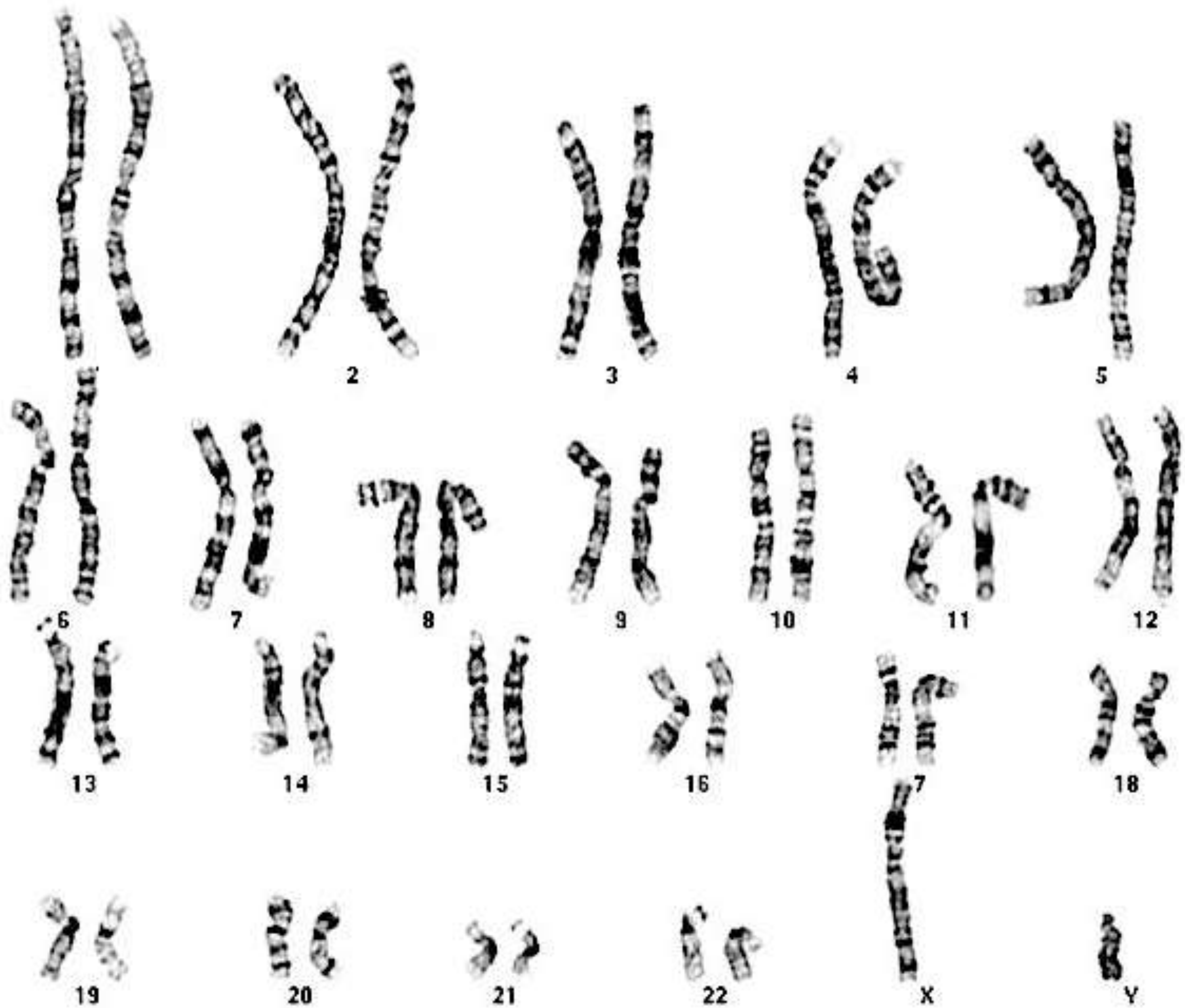
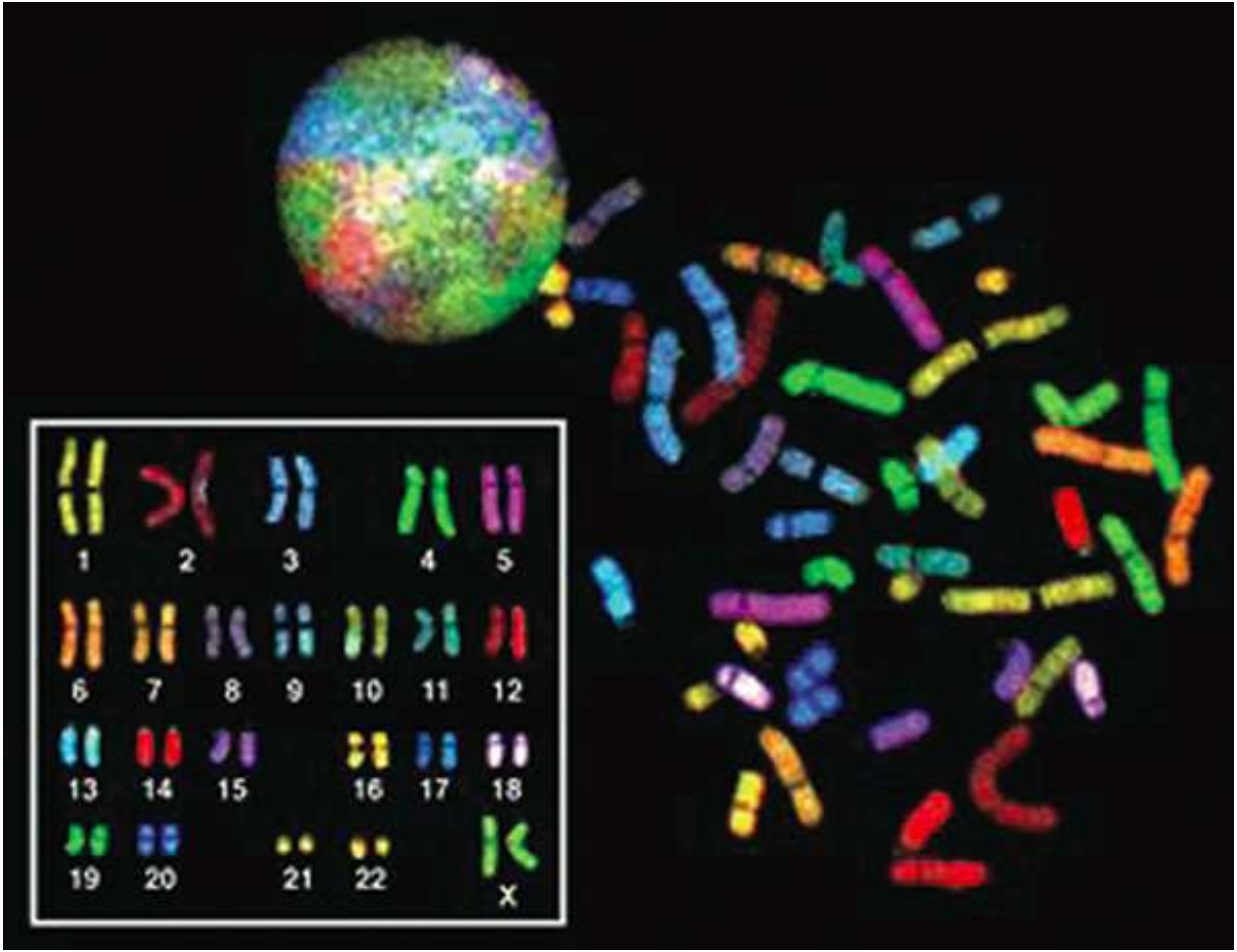
Sreelaja Nair

Karyotyping

- Done by isolating metaphase DNA from nuclei of cell
- DNA is stained with a dye, results in a pattern of bands on the chromosome = G banding
- Stained chromosomes are lined up on a slide in pairs and imaged
- Each chromosome has a unique G-banding pattern



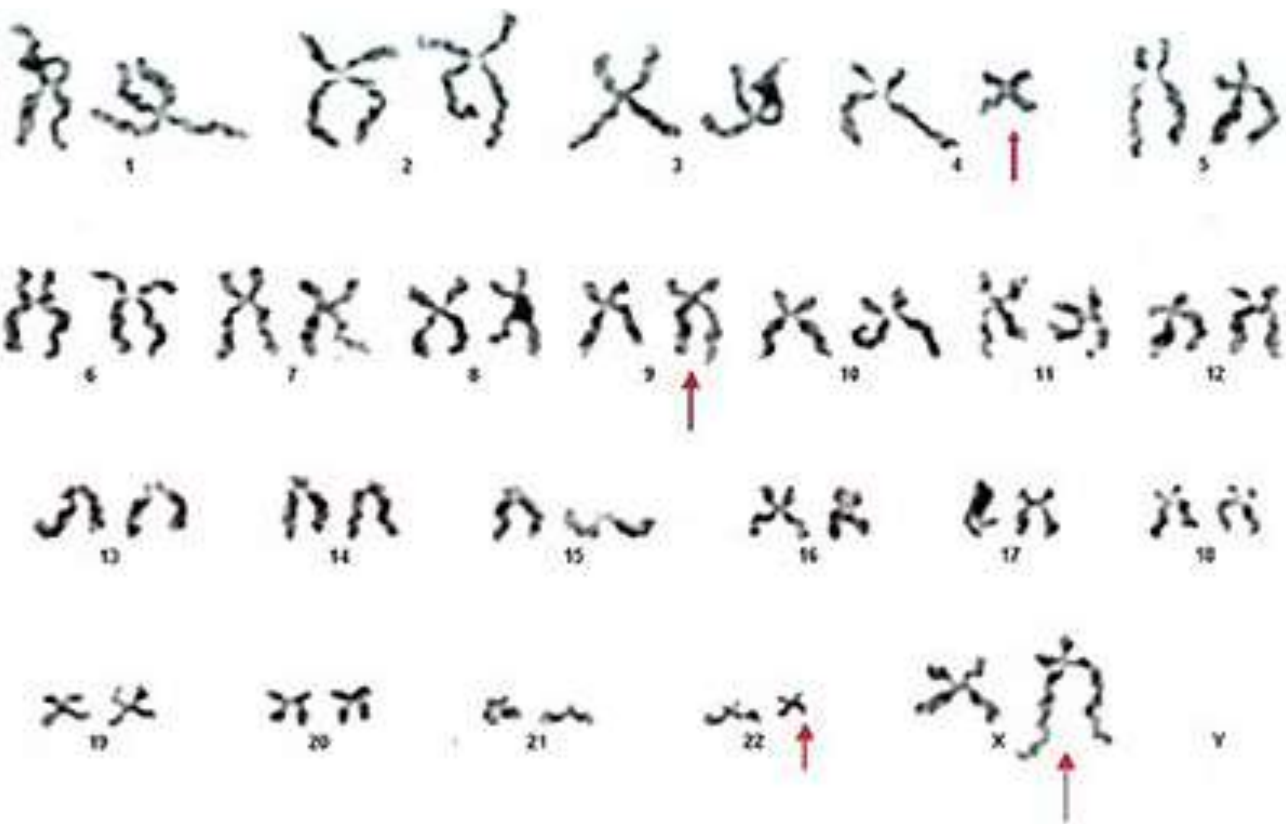
Spectral karyotyping

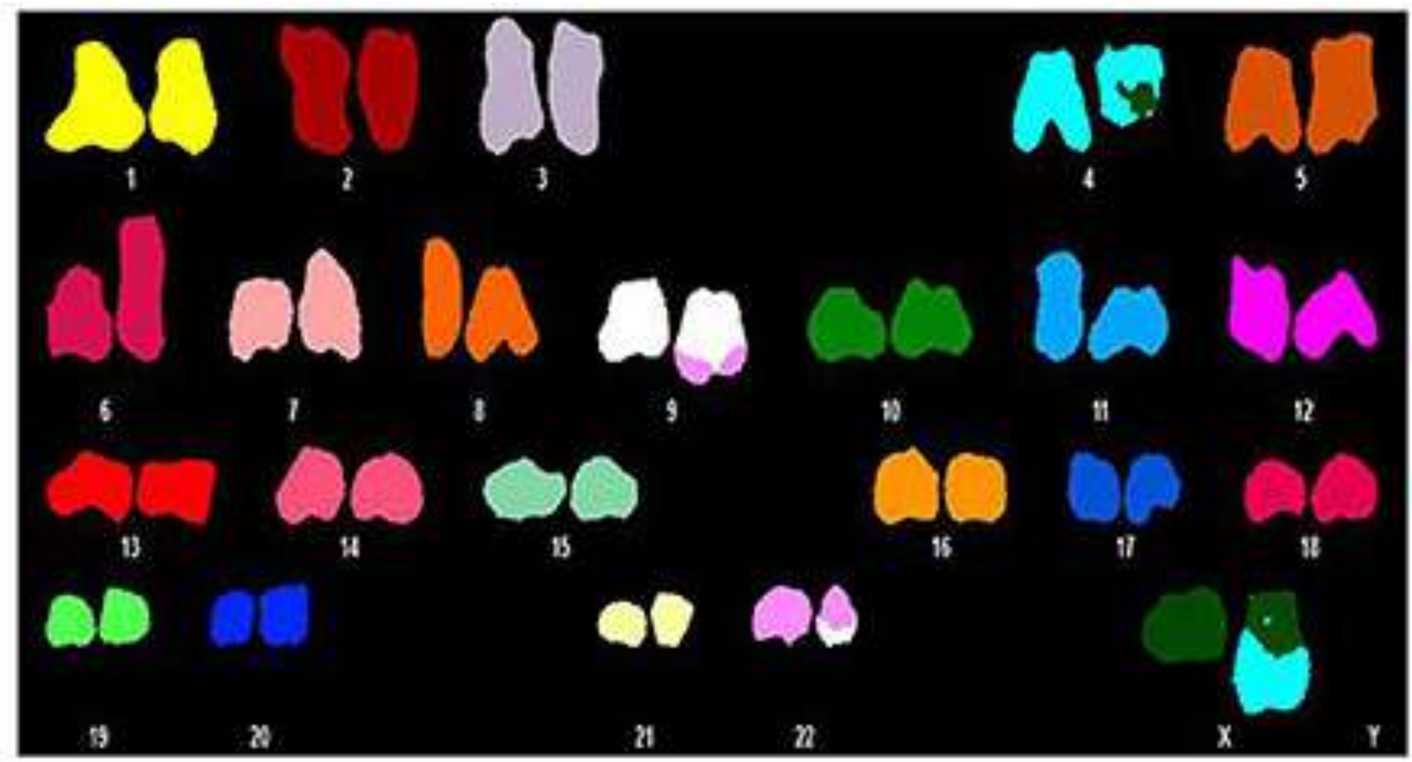


https://www.cytotest.com/sky.asp

Spectral karyotyping

Makes it visually easy to identify chromosomal structural abnormalities





When chromosomes break and rejoin, genes “break” and “fuse”.

Causes loss of function or creates “fusion genes” which have abnormal function

PMID: 30891424

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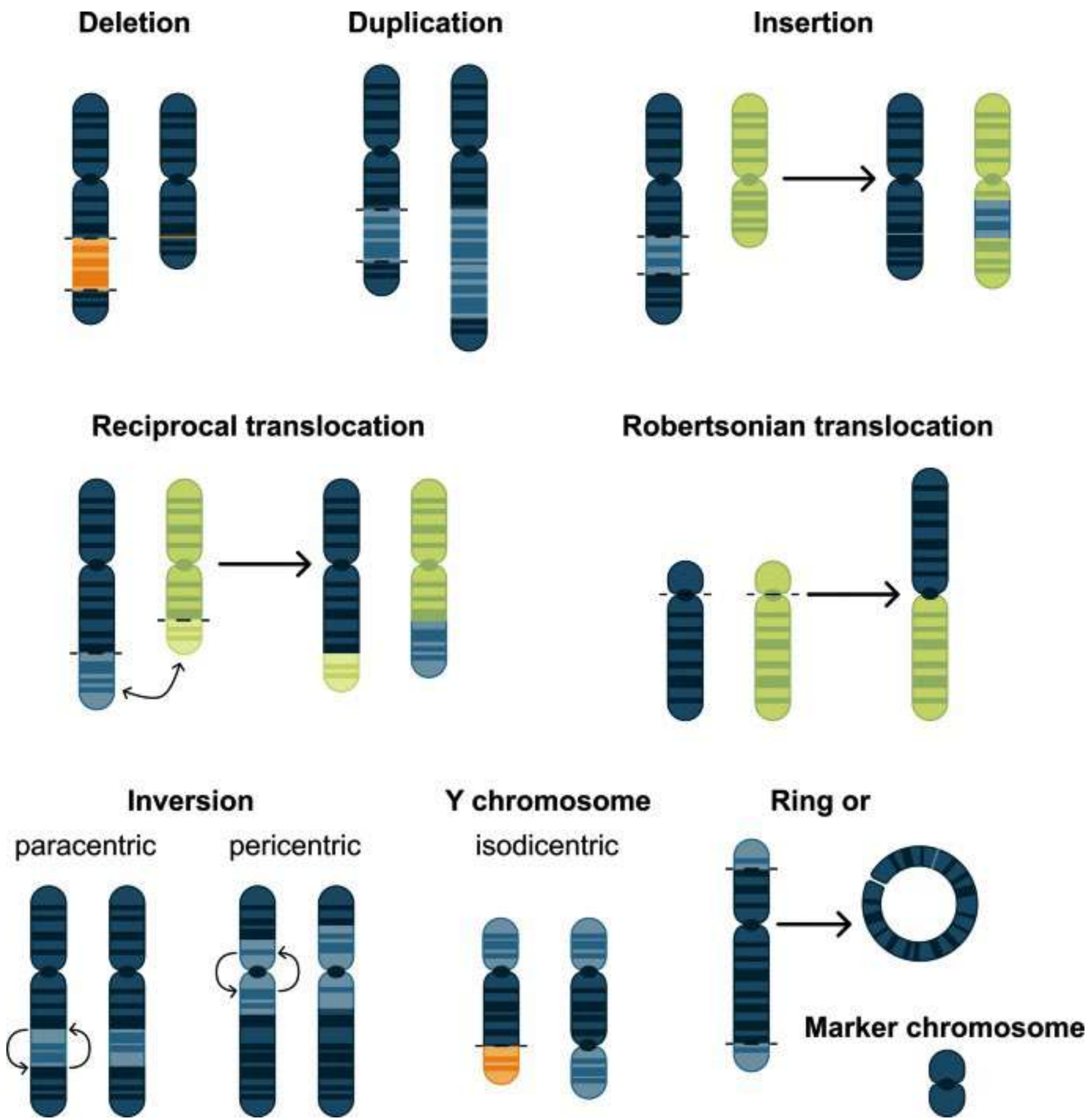
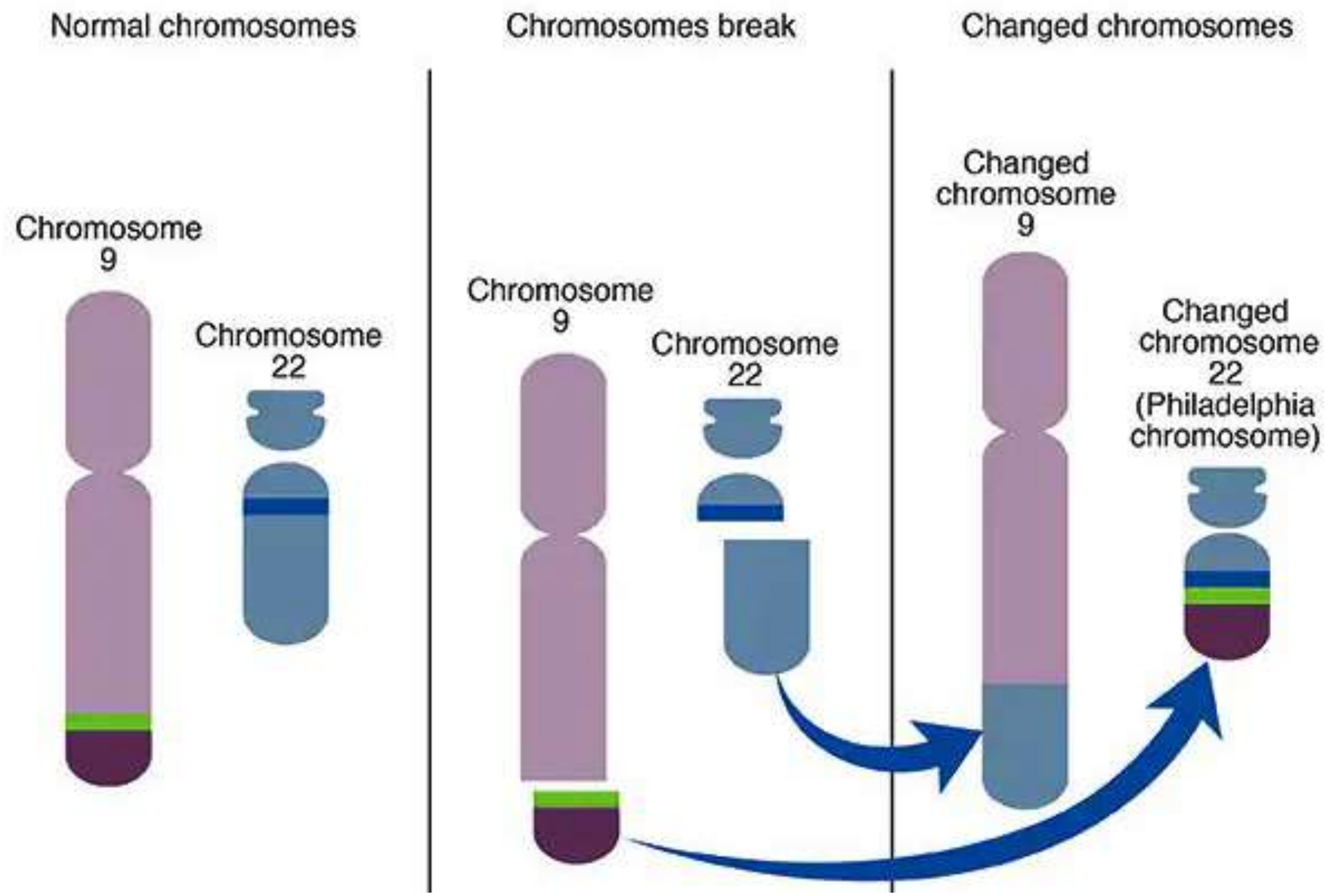
Structural abnormalities of chromosomes



https://doi.org/10.1007/978-3-031-31574-9_23

Figure 15.14 and 15.15 of Campbell's Biology: a global approach

Philadelphia Chromosome

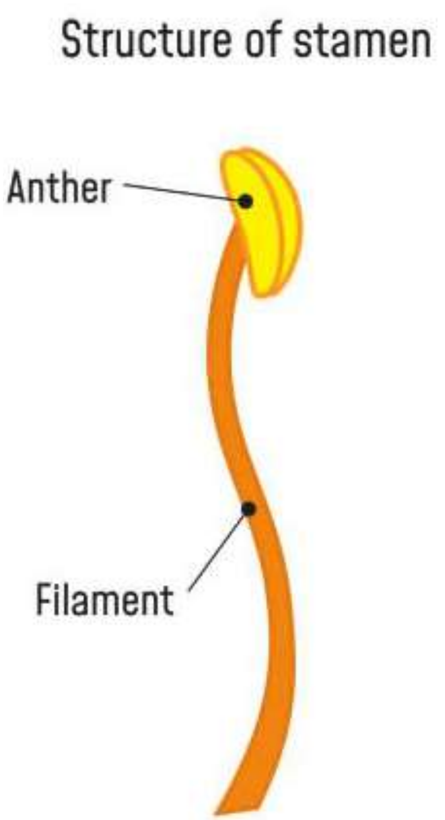


- Translocation between chromosomes 9 and 22
- Creates a longer chromosome 9
- Moves the gene *abl* on chromosome 9 next to *bcr* on chromosome 22 creating a fusion gene *bcr-abl* on 22
- b*cr-abl* is an oncogene
- Oncogene = genes whose activity cause cancer

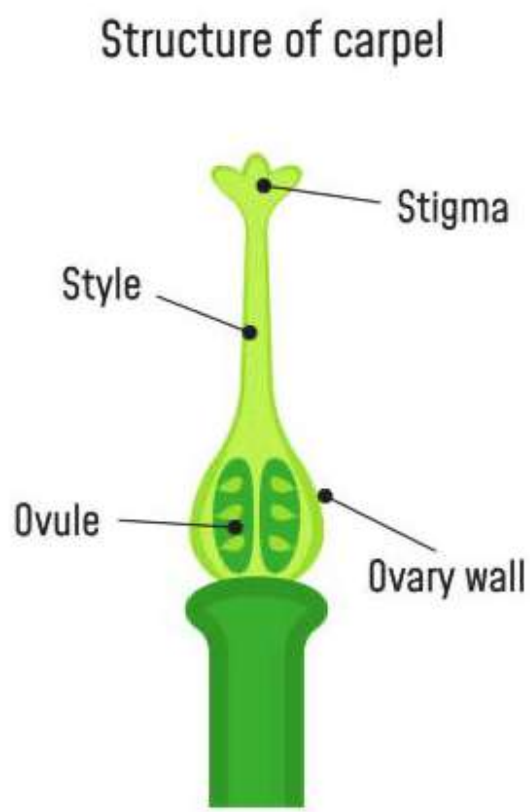
Figure 15.16 of Campbell's Biology: a global approach

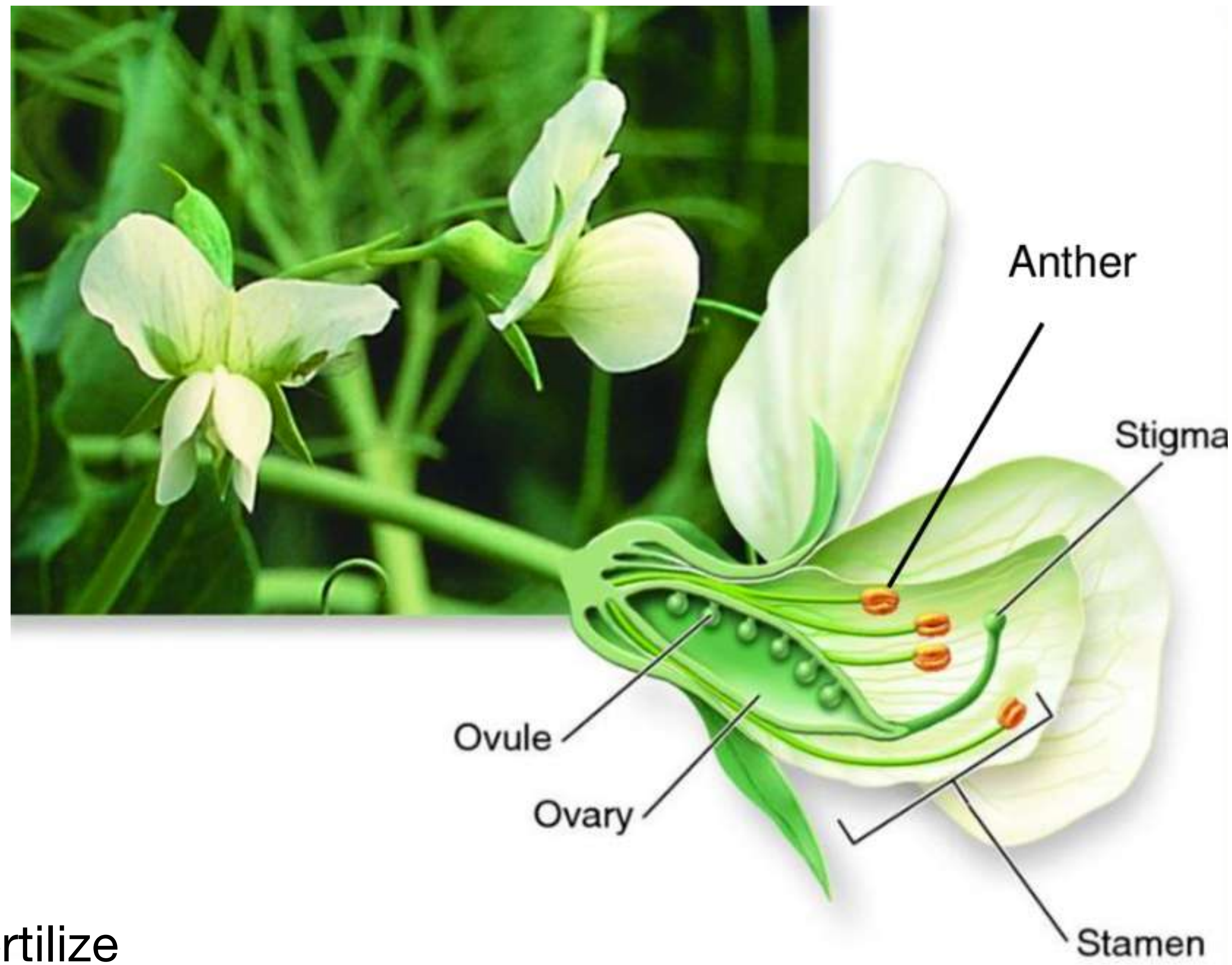
Gregor Mendel and the Pea plant

Male



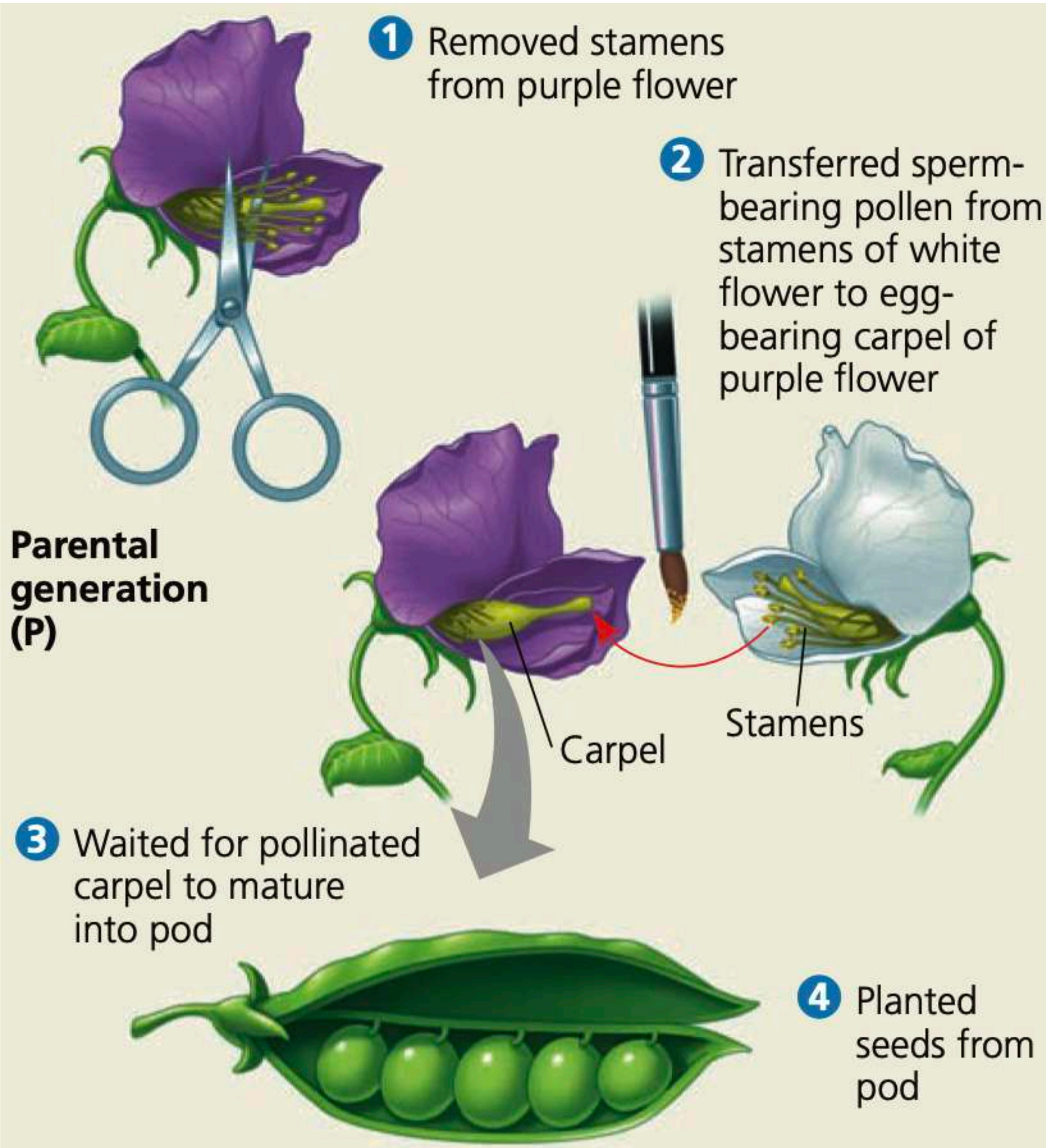
Female



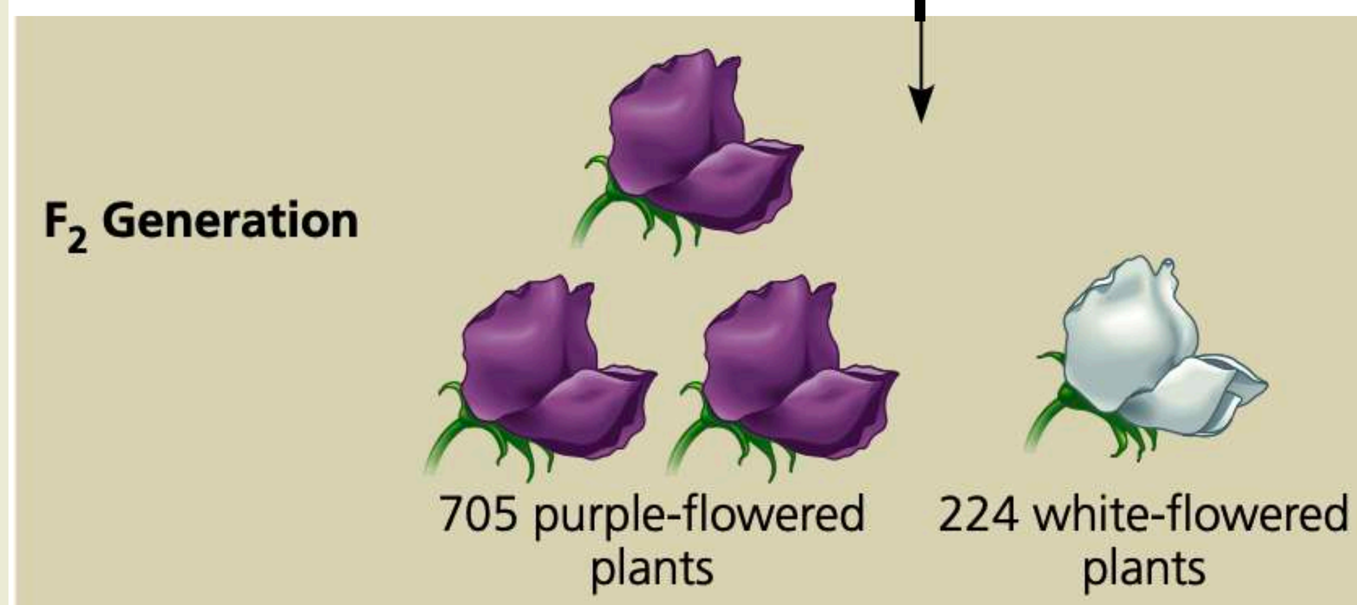


- Plants can self-fertilize or cross-fertilize
- Can take pollen from anther of one flower and dust it on the stigma of another flower
- Technical advantage for experimental mating (crossing) of plants

Gregor Mendel and the Pea plant



Self or cross-fertilize F1 generation



The result was the same for the reciprocal cross, which involved the transfer of pollen from purple flowers to white flowers

Figure 14.2 of Campbell's Biology: a global approach

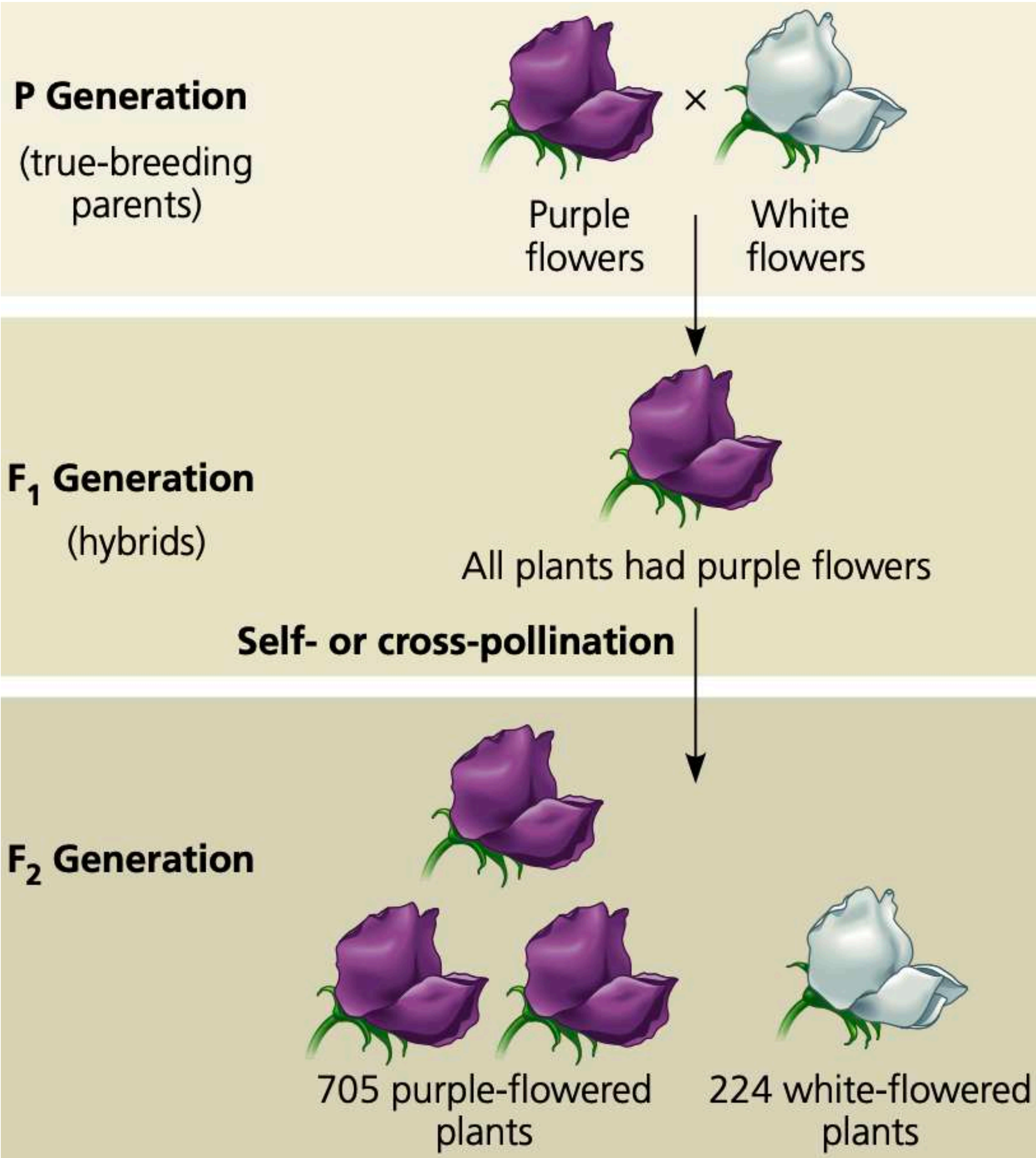
Gregor Mendel and the Pea plant

- Mendel chose to track only those characters that occurred in two distinct, alternative forms, such as purple or white flower color
- He started his experiments with varieties that, over many generations of self-pollination, had produced only the same variety as the parent plant = true breeding
- The mating, or cross-pollinating (crossing) of two true-breeding varieties is called hybridization
- The true-breeding parents are referred to as the P generation (parental generation), and their hybrid offspring are the F1 generation (first filial generation, the word filial from the Latin word for "son").
- Allowing these F1 hybrids to self-pollinate (or to cross-pollinate with other F1 hybrids) produces an F2 generation (second filial generation).

- Mendel's quantitative analysis of the F2 plants from thousands of genetic crosses like these allowed him to deduce two fundamental principles of heredity, which have come to be called the law of segregation and the law of independent assortment

Sreelaja Nair

Law of Segregation



Crossed two true-breeding plants that each produce Purple or White flowers

When seeds from F1 generation grew into plants, all plants produced Purple flowers. What happened to the information to produce White flowers?

When F1 generation self or cross-fertilized to produce seeds and when those seeds grew into plants, some of F2 plants produced White flowers

White flowers in F2 plants did not appear randomly, they appeared in a ratio of 3:1 (3 purple flower plants:1 White flower plant)

Figure 14.3 of Campbell's Biology: a global approach

Law of Segregation

- Mendel reasoned that the heritable factor for white flowers did not disappear in the F1 plants but was somehow hidden, or masked, when the purple-flower factor was present.
- In Mendel's terminology, purple flower color is a dominant trait, and white flower color is a recessive trait.
- The reappearance of white-flowered plants in the F2 generation was evidence that the heritable factor causing white flowers had not been diluted or destroyed by coexisting with the purple-flower factor in the F1 hybrids.
- Instead, it had been hidden when in the presence of the purple flower factor.
- Mendel observed the same pattern of inheritance in six other characters, each represented by two distinctly different traits

Law of Segregation

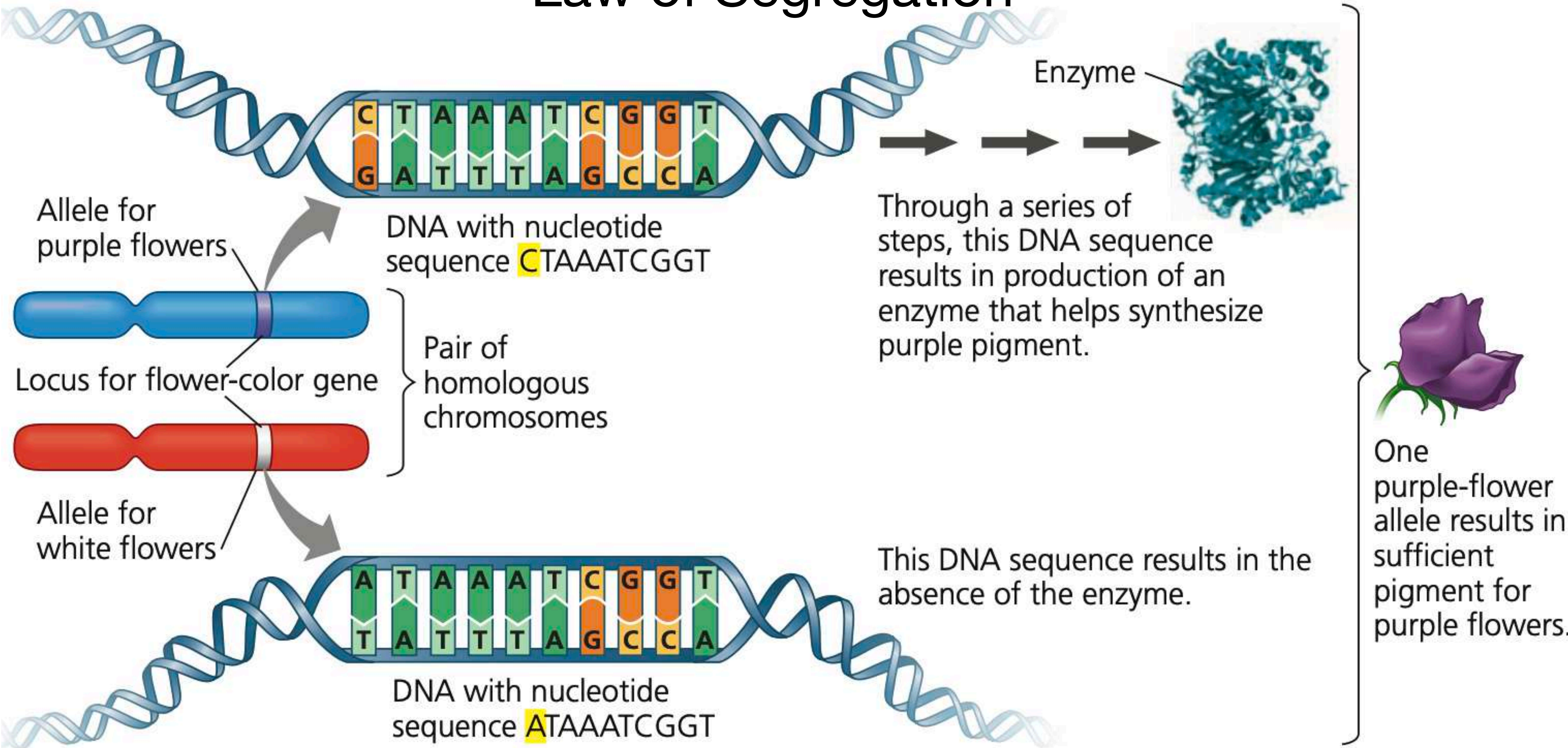
Table 14.1 The Results of Mendel's F_1 Crosses for Seven Characters in Pea Plants

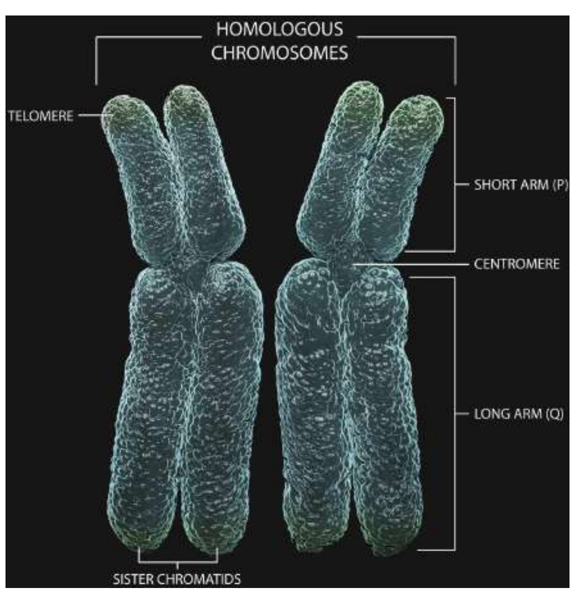
Character	Dominant Trait	×	Recessive Trait	F_2 Generation Dominant: Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

1. Alternative versions of genes account for variations in inherited characters.
2. For each character, an organism inherits two copies (that is, two alleles) of a gene, one from each parent.
3. If the two alleles at a locus differ, then one, the dominant allele, determines the organism's appearance; the other, the recessive allele, has no noticeable effect on the organism's appearance.
4. Law of segregation, states that the two alleles for a heritable character segregate (separate from each other) during gamete formation and end up in different gametes.

Table 14.1 of Campbell's Biology: a global approach

Law of Segregation

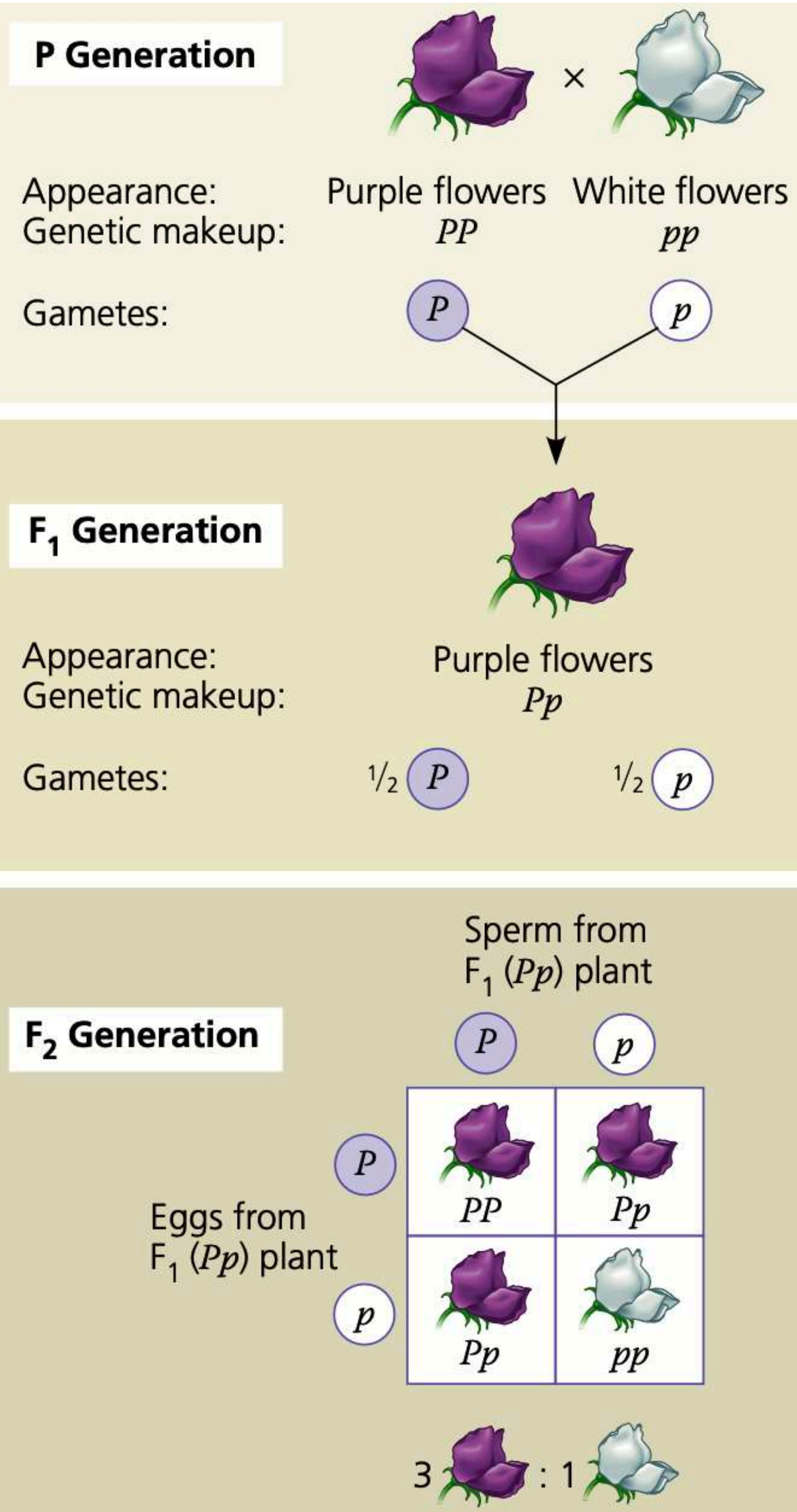




- Alleles = alternative versions of a gene
- For a diploid cell at metaphase, there will be 4 alleles of a gene - one on each sister chromatid
- A haploid gamete has one chromatid at the end of meiosis = one allele
- When two haploid gametes fuse (fertilization) - embryo gets two alleles, one from each gamete

Figure 14.4 of Campbell's Biology: a global approach

Law of Segregation



- Each true breeding P plant has one allele of the gene, either *P* or *p*. Each P plant has two copies of the allele (*PP* or *pp*) as it is a diploid
- Each P plant produces one type of haploid gamete, either, Purple allele gamete (*P*) or White allele gamete (*p*)

- When gametes from two P plants fuse, they can only form one diploid combination in F1 upon fertilization: Pp
- The cross-pollinated F1 plant has two alleles *P* and *p*, but P is dominant and therefore F1 plant has Purple flowers
- Each F1 plant produces two types of gametes: *P* and *p* = segregation

- When gametes of F1 plants fuse (self or cross-fertilization), they can form three types of diploid combinations in F2: PP or Pp or pp.
- PP = Purple flower F2 plant = like one type of P generation
- Pp = Purple flower F2 plant = like the F1 generation
- pp = White flower F2 plant = like the second type of P generation, reappearing after skipping one generation

Figure 14.5 of Campbell's Biology: a global approach